

Design FMEA (qualitative version, based on simulation signals)

Qualitative S/O ratings are three-level (high/medium/low) based on simulation-signal magnitude vs threshold. Complete FMEA (process/supplier) is N/A (beyond pure-simulation boundary).

Failure Cause	Simulation signal (detectability basis)	Severity	Occurrence	Mitigation (implemented in design)
Electrode drives anode potential < 0 V (tight N/P + concentration polarization)	<code>`anode_potential_v` < 0`</code>	High	Low	anode 108 μm (N/P 1.04), 1.5 μm anode passivation
Electrode + 1C discharge reaction heat at 45 °C ambient	<code>`T_max_K` vs 323.15 K</code>	High	Low	active cooling h=400 W/m²·K → T_max 321.68 K
Electrode / low active loading	<code>1C`capacity_ah` + `energy_density_wh_l`</code>	Medium	Low	6.372 Ah, 1260 Wh/L (≥ 950)
Open-Circuit Voltage (OCV) below 4.1 V	<code>`midpoint_voltage_v`</code>	Medium	Low	LNMO 4.7 V cathode → 4.131 V
Electrode drives anode more negative, accelerating SEI	<code>`sei_thickness_nm_end`</code>	Medium	Low	artificial-SEI coating (SEI k 2e-15 m/s) → 19.5 nm
Specificity 4e-15 m²/s (NMC value inherited)	4C CC charge reaches ~27 % SOC	Medium	Medium	noted honestly; not resolvable within allowed time

Conclusion

- Highest-risk items: thermal runaway risk and negative-electrode plating — both mitigated and verified (T_max 321.68 K, no plating).
- Remaining residual risk: positive-electrode diffusion limitation caps 4C CC charge depth (~27 % SOC); full FMEA (process/supplier) is **N/A (beyond pure-simulation boundary)**.